

ONLINE NIGHT 1 · 5:30 PM – 8:00 PM CST

Pregnancy Physiology

& When Pregnancy Goes Wrong

Mary's Hands Network
ICEA Certified Birth Doula
Hybrid Training

Learning Objectives

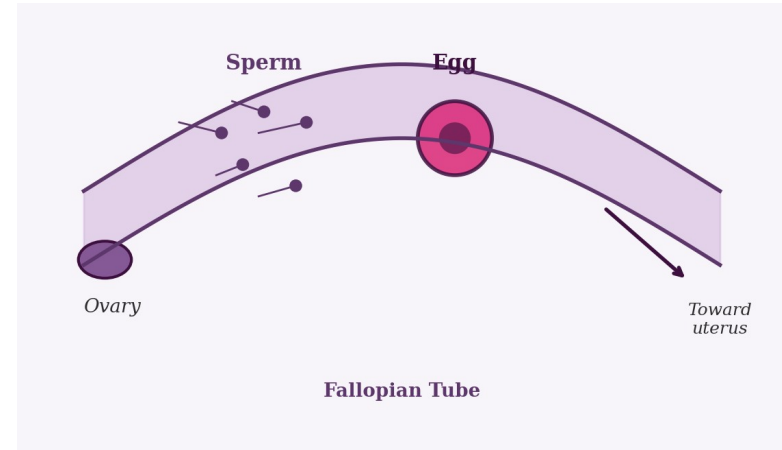
- Pregnancy by trimester — physical & emotional
- Common discomforts — evidence-based comfort
- Prenatal testing + doula's prenatal role
- Recognize preeclampsia & complications
- Advocacy + Louisiana health disparities

Normal Pregnancy:

Physiology & the Doula's Role

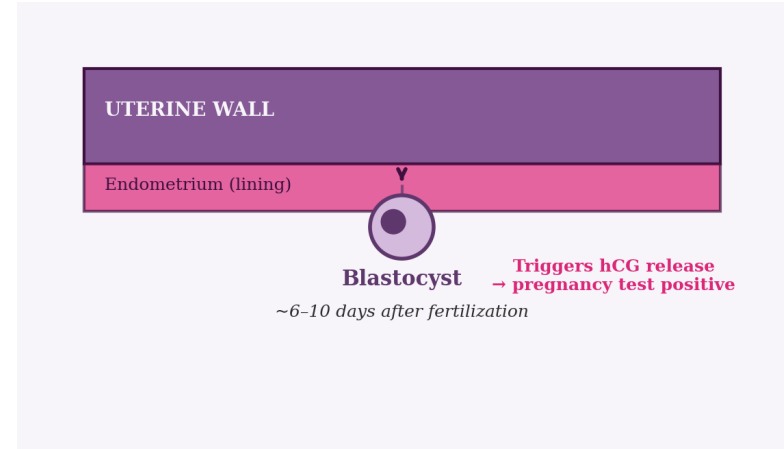
Fertilization

- Sperm meets egg in fallopian tube
- Zygote begins dividing immediately
- Genetic blueprint fixed at conception
- Pregnancy is dated from last period
- Many are 2 weeks "pregnant" already



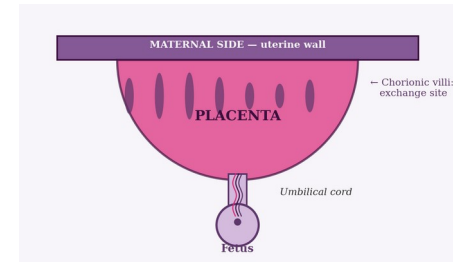
Implantation

- 6–10 days after fertilization
- Blastocyst burrows into uterine lining
- Triggers hCG — pregnancy test positive
- Some notice light spotting
- If it fails, period follows



The Placenta

The only organ the human body grows — and then discards.



Delivers Oxygen & Nutrients

Maternal and fetal blood never directly mix. The placenta acts as a selective barrier — passing oxygen, glucose, and vitamins through to the fetus while blocking most pathogens and toxins.

Removes Fetal Waste

Carbon dioxide and metabolic waste products from the fetus cross back through the placenta into the mother's circulation for disposal.

Produces Key Hormones

Takes over hormone production around week 10 — producing hCG, progesterone, and estrogen to sustain the pregnancy and drive physical changes.

Development & Delivery

Fully functional by weeks 12–16. Continues growing until birth, ultimately weighing about 1–1.5 lbs. Delivered as the third stage of labor.

Six Hormones Drive Every Symptom Your Client Feels

Understanding these hormones lets you explain exactly why your client feels the way she does.

hCG

Causes nausea, detected by pregnancy tests, peaks at week 10

Progesterone

Maintains the uterine lining, causes heartburn, fatigue

Estrogen

Drives growth, blood flow, skin changes, the pregnancy glow

Relaxin

Loosens ALL ligaments — causes back pain, SPD, clumsiness

Oxytocin

Builds through pregnancy, triggers labor, drives bonding

Cortisol

Rises throughout pregnancy — chronic stress elevation = risk

hCG — Human Chorionic Gonadotropin

The hormone that tells the body: pregnancy is happening.

- Produced by the placenta
- Doubles every 48–72 hours early on
- Peaks around week 10
- Triggers the nausea center
- Pauses ovulation for the pregnancy

Progesterone — "The Pregnancy Hormone"

Without progesterone, the uterus cannot sustain the pregnancy.

- Maintains the uterine lining
- Suppresses contractions, protects pregnancy
- Relaxes smooth muscle everywhere
- Sedating — drives T1 fatigue
- Sudden postpartum drop → baby blues

Estrogen — Growth, Blood Flow & Change

Estrogen drives visible physical transformation — from the glow to the stretch marks.

- Drives uterine growth
- Develops breast tissue for lactation
- Creates the "pregnancy glow"
- Also: nasal congestion, bleeding gums
- Stretch marks, linea nigra, melasma

Relaxin — The Joint & Ligament Relaxer

It loosens the pelvic ligaments for birth — and every other ligament along with them.

- Loosens pelvic ligaments for birth
- Loosens ALL ligaments — not just pelvic
- Back pain, SPD, round ligament pain
- Causes "pregnancy clumsiness"
- Resolves after breastfeeding ends

Oxytocin — "The Love Hormone"

Your calm, warm, reassuring presence as a doula is not just emotional support — it is hormonal support.

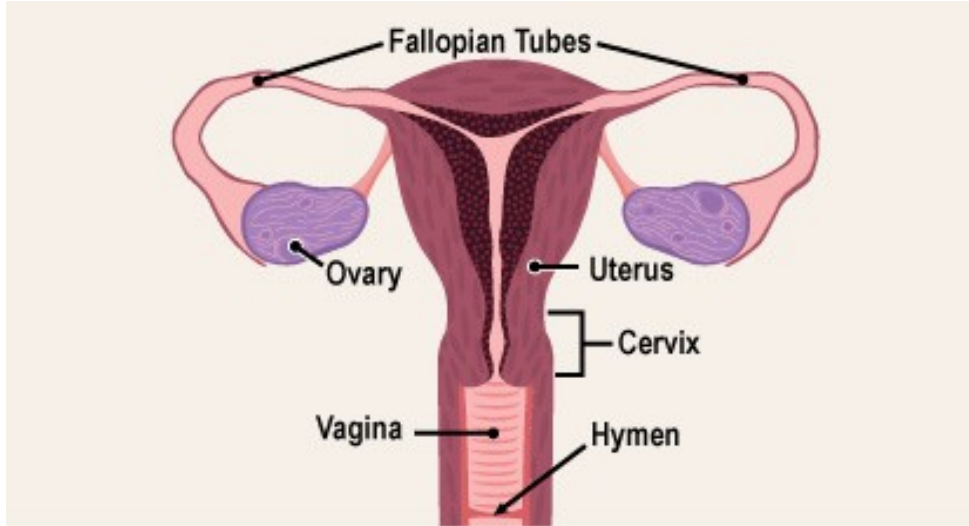
- Builds gradually, peaks at labor
- Drives contractions (Pitocin is synthetic)
- Responds to touch, eye contact, safety
- Powers skin-to-skin bonding
- Triggers milk let-down reflex

Cortisol — The Stress Hormone

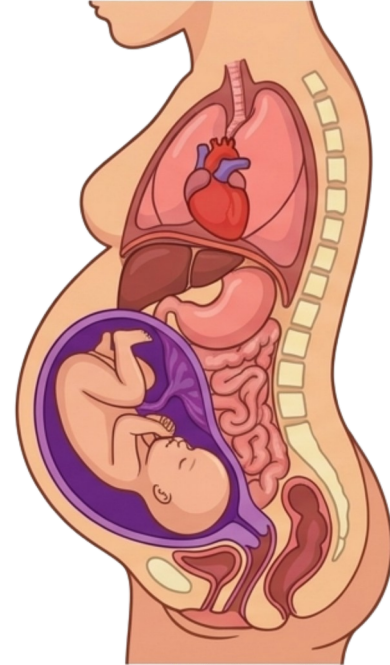
Rising cortisol in pregnancy is normal. Chronic stress-driven elevation is where doulas make a difference.

- Rises throughout pregnancy — normal
- Adapts immune system to tolerate fetus
- Supports fetal lung development
- Chronic elevation → preterm risk
- Doula presence measurably lowers it

Where Does Everything Go?



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The Uterus — The Engine of It All

From pear-sized to watermelon-sized. Everything else moves to make room.

- Pear-sized, 3 oz pre-pregnancy
- Watermelon-sized, ~2 lbs at term
- Pushes every organ upward and aside
- Drives most T3 symptoms
- One of the body's strongest muscles



Stomach, Intestines & Diaphragm

Stomach & Intestines

T2 onward, worst in T3

Pushed upward and compressed by the growing uterus.

Progesterone slows gut motility. Result: heartburn, constipation, bloating, and slower digestion. Heartburn worsens after meals and when lying flat.

Diaphragm

Third trimester

Pushed upward by about 4 cm in T3, reducing lung capacity.

Causes shortness of breath — especially lying flat or climbing stairs. Most clients feel relief when the baby drops (lightening) in late T3.

Bladder, Kidneys & Cardiovascular System

Bladder

T1 and T3

Compressed by hCG in T1 and by the baby's head in T3. Frequent urination is entirely structural. Clients who reduce fluids to cope risk dehydration and UTIs.

Kidneys

Throughout pregnancy

Filtration rate increases 60–80% to handle expanded blood volume. Mild glucose in urine (glycosuria) is normal — the kidneys are simply filtering more than usual.

Heart & Circulation

T1 through birth

Blood volume increases 40–50%. Heart rate rises 10–20 bpm. Cardiac output increases 30–50%. This explains fatigue, dizziness on standing, and the swelling that comes from fluid redistribution.

BREAK

6:20 PM — 6:30 PM · 10 minutes

Step away, stretch, hydrate, reset.

Back at 6:30 PM — trimesters & discomforts up next.

The Foundation — Critical Development

- Day 28: neural tube closes
- Weeks 5–8: all major organs form
- Week 8: now a fetus — fingers form
- Week 10: placenta takes over
- Week 13: 3 inches, facial features



Week 4
~poppy seed



Week 8
~raspberry



Week 13
~lemon · 3 in

Often the Hardest — With No Visible Bump

- Nausea (70–80%) — not just morning
- Extreme fatigue — building a placenta
- Heightened smell, food aversions
- Pregnancy feels surreal — no bump yet
- Highest miscarriage risk — anxiety work

The Growth Surge — Senses & Viability

- Week 16: fingerprints form
- Weeks 18–22: quickening — first kicks felt
- Week 20: swallows, tastes, hears
- Week 24: viability threshold
- Week 27: eyes beginning to open



Week 16
~avocado



Week 20
~banana · 10 in



Week 24
~corn · viability



Week 27
~cauliflower · 2 lb

The Honeymoon Period — Energy Returns

- Nausea fades, energy returns
- Baby bump becomes visible
- Round ligament pain (sharp belly)
- Skin changes: glow, stretch marks, melasma
- Prime relationship-building time



Prime prenatal relationship-building time.

The Final Preparations — Fat, Lungs & Position

- Weeks 28–32: rapid weight gain
- Weeks 32–36: surfactant produced
- Week 37: "early term" — brain still growing
- Weeks 39–40: developmentally optimal
- Full term: 19–21 in, 6–9 lbs



Week 28

~eggplant · 2.5 lb



Week 32

~squash · 4 lb



Week 36

~papaya · 6 lb



Week 40

~watermelon · 7-8 lb

The Home Stretch — Intensity & Anticipation

- Discomfort, poor sleep, breathlessness
- Heartburn intensifies
- Pelvic pressure, "lightning crotch"
- Swelling: sudden severe = red flag
- Nesting + emotional intensity



Anticipation, fear, excitement, impatience — all at once.

Nausea & Vomiting

Affects 70–80% of pregnant people. Can happen any time of day.

What you can offer:

- Small, frequent meals
- Ginger — tea, candies, or fresh
- Vitamin B6 (with provider OK)
- Sea-Bands (acupressure)
- Avoid known triggers

When to refer to provider:

Inability to keep fluids down for 24 hours, weight loss, or vomiting more than 3–4 times per day may indicate hyperemesis gravidarum — refer to provider.

Doula script:

"Morning sickness is miserable and I'm sorry you're dealing with it. Try eating small amounts frequently — even crackers or toast. Protein helps. And ginger really works for some people."

Heartburn & Indigestion

Up to 80% of pregnant people experience this — especially in the third trimester.

What you can offer:

- Small, frequent meals
- Avoid spicy, acidic, fatty foods
- Don't lie down 2–3 hrs after eating
- Sleep with head elevated
- Chew gum to neutralize acid

When to refer to provider:

If severe or not responding to dietary changes, the provider can recommend safe antacids for use during pregnancy.

Doula script:

"Heartburn is so common in pregnancy because hormones relax the valve at the top of your stomach and the baby is pushing everything upward. Small meals and sleeping propped up really help."

Swelling (Edema)

Normal in the feet, ankles, and hands — especially in the third trimester. Blood volume has increased 40–50%.

What you can offer:

- Elevate feet when resting
- Avoid prolonged standing
- Stay hydrated — reduces swelling
- Reduce sodium intake
- Compression stockings if needed

When to refer to provider:

Sudden, severe swelling in the face, hands, or around the eyes — especially with headache or vision changes — requires immediate evaluation. One leg significantly more swollen than the other may indicate a blood clot.

Doula script:

"Swelling in your feet and ankles is really common, especially in the heat or at the end of the day. Elevating your feet helps. Drinking water actually helps reduce it, even though that seems backward."

Back & Pelvic Pain

Caused by shifted center of gravity, relaxin loosening ligaments, and a growing uterus.

What you can offer:

- Posture: shoulders back, pelvis tucked
- Supportive shoes — avoid heels
- Prenatal yoga, gentle stretching
- Heat for muscle pain (not belly)
- Side-lying with pillow between knees

When to refer to provider:

Severe, persistent pain or symphysis pubis dysfunction (SPD) may benefit from a referral to a prenatal physical therapist.

Doula script:

"Your body is doing a lot of shifting and adjusting. Back pain is really common. Stretching helps, heat helps, and changing positions often helps. If it gets bad, prenatal PTs do amazing work."

Fatigue

Real, intense, and often underestimated — especially in T1 and again in T3.

What you can offer:

- Permission to rest — not laziness
- Short naps (20–30 minutes)
- Protein + complex carbs for energy
- Hydration — dehydration worsens fatigue
- Gentle movement (walking) helps

When to refer to provider:

Extreme, debilitating fatigue beyond what is typical may indicate anemia — encourage provider to check iron levels.

Doula script:

"Your body is working so hard right now — harder than it would running a marathon. Of course you're tired. Rest when you can. This isn't laziness. This is your body doing exactly what it needs to do."

Braxton Hicks Contractions

Practice contractions — irregular, usually painless, and stop with position change or hydration.

What you can offer:

- Hydration first — common trigger
- Position change often stops them
- Rest if frequent
- Reassurance: uterus practicing, not labor

When to refer to provider:

If contractions become regular (every 5–10 minutes), are accompanied by pain, bleeding, or pelvic pressure before 37 weeks — call the provider. When in doubt, get it checked out.

Doula script:

"Those are Braxton Hicks contractions — your uterus is practicing for labor. They're normal and actually helpful. But if they become regular or painful before 37 weeks, always check in. When in doubt, get it checked out."

Prenatal Testing — What Your Clients Go Through

Test	Timing	What It Screens	Your Role
First Trimester Screening	10–13 wks	Chromosomal risk assessment. Blood + nuchal translucency ultrasound. A risk estimate — NOT a diagnosis.	<i>Help clients understand a high-risk result is not a diagnosis. Many lead to healthy babies.</i>
Anatomy Scan	18–22 wks	Detailed fetal anatomy, placenta, fluid. Often reveals biological sex if parents want to know.	<i>Normalize both the excitement and the anxiety. Help them write down questions beforehand.</i>
Glucose Screening	24–28 wks	Screens for gestational diabetes. A high 1-hour result does not automatically mean GDM.	<i>Encourage bringing a snack for after. Normalize the test.</i>
Group B Strep	36–37 wks	Normal bacteria in 25–30% of pregnant people. GBS+ = IV antibiotics during labor.	<i>Normalize the positive result. Not an infection. Not from anything they did wrong.</i>

Prenatal Vitamins — Why They Matter

Folic acid prevents neural tube defects — which occur in the first 28 days, often before the person knows they are pregnant. Supplementation reduces the risk by 50–70%. Iron is equally critical — blood volume expands 40–50% and iron deficiency affects up to 52% of pregnant people globally.

When clients can't or won't take them:

- "They make me nauseous." Try taking them at night, with food, or switch to gummies (note: gummies often lack iron)
- "I can't afford them." WIC provides free prenatal vitamins. Community health centers and OB offices often have samples
- "I keep forgetting." Set a daily phone alarm. Keep them next to something used every day

"Pregnancy Brain" Is Real — And Adaptive

2016 Nature Neuroscience study

Brain remodels: social cognition, empathy

Persists 2+ years postpartum

Strengthens maternal-infant bonding

Short-term memory dips — adaptive trade-off

Doula Script

"Your brain is literally rewiring itself right now to help you bond with and care for your baby. Research shows these changes are actually preparing you for parenthood. So yes, you might forget where you put your phone — but your brain is getting really good at reading your baby's cues. It's a trade-off, and it's temporary."

Scope of Practice — Knowing the Line

Within Your Scope

- Provide accurate information about normal pregnancy
- Help clients formulate good questions for their providers
- Recognize when symptoms need medical evaluation
- Support clients through normal discomforts
- Normalize prenatal testing and monitoring
- Support birth partners in their evolving role

Outside Your Scope

- Diagnosing any condition or symptom
- Recommending specific medications or treatments
- Interpreting lab results or test findings
- Advising clients to delay seeking medical care
- Telling clients whether to accept or decline interventions
- Overriding or contradicting medical advice

Imagine a client at 36 weeks who has been Googling everything.

Based on what you just learned, what kinds of questions do you think she might bring to a prenatal visit? Pick 2–3 your group comes up with and work through how you would respond — and where scope of practice starts to matter.

Each group shares with the full class:

- What is the question?
- What would you say?
- What would you NOT say — and why?

BREAK

7:20 PM — 7:30 PM · 10 minutes

Step away, stretch, hydrate, reset.

Back at 7:30 PM — When Pregnancy Goes Wrong.

When Pregnancy Goes Wrong:

Danger Signs & Complications

Your job is not to diagnose. Your job is to recognize, advocate, and act.

Scope of Practice Still Applies Here

Important — Scope Reminder

You are not a medical provider. You do not diagnose. You do not prescribe. You do not override medical advice.

But you ARE an educated support person who knows when something is not right.

Sometimes that means calmly and clearly saying:

"We need to call your provider right now."

That is not overstepping. That is exactly what doulas do.

Preeclampsia

5-8%
of all
pregnancies

What It Is

High blood pressure ($\geq 140/90$) developing after 20 weeks of pregnancy, with signs of organ stress — protein in urine, liver dysfunction, kidney problems, or low platelets.

Clinical Note

- Severe features: BP $\geq 160/110$
- Only cure is delivery
- HELLP variant: 10–20% of severe cases
- Can develop after birth (up to 6 wks)

Every doula must know these.

01



Severe Persistent Headache

Not responding to Tylenol

02



Vision Changes

Blurred, spots, flashing lights

03



Upper Abdominal Pain

Especially on the right side

04



Sudden Severe Swelling

Face, hands, around the eyes

05



New Nausea or Vomiting

Onset in the third trimester

06



Decreased Fetal Movement

Always take seriously

What Happens After Diagnosis

Treatment

- Only cure is delivery
- Magnesium sulfate prevents seizures
- Antihypertensives prevent stroke
- Monitoring: BP, labs, NST, BPP

Postpartum Risk

- 1 in 4 eclamptic seizures postpartum
- Most occur in first 48 hours
- Can develop up to 6 weeks postpartum
- Many had no symptoms during pregnancy

Doula Scripts — Advocacy in Practice

When a provider minimizes symptoms:

"I understand that might be a normal variation, but she's really concerned, and these are classic warning signs. Could we check her blood pressure and urine just to give everyone peace of mind?"

When the headache has lasted all day:

"She's saying this is the worst headache of her life and it's not responding to Tylenol. Given her history, could we have her evaluated today?"

When your client hesitates to call:

"I know you don't want to bother them — but this is exactly what they're there for. It's better to check and have it be nothing. Let's call together."

The disparity is not a coincidence — it is a system.

2-3x

more likely to die

Black women face 2–3x higher pregnancy-related mortality than white women nationally

60%

more likely to develop preeclampsia

Higher rates and worse outcomes when they do

2x

Louisiana's maternal mortality rate

Nearly double the national average — gap is most severe for Black mothers

Clinical Note

This disparity holds across income and education levels — structural racism, not individual risk, is the primary driver. Symptoms are more often minimized in Black patients. Your informed, persistent advocacy is not optional — it is a clinical intervention.

Where have you seen, read about, or heard stories of racism in healthcare?

In the news. In your own family. In your community. In your own experience as a patient.

How does knowing this change what you think a doula's role should look like?

There are no wrong answers. No requirement to have a birth work story to contribute. Your instructor will hold space for whatever comes up.

- What surprised you most about the statistics?
- What would it look like for a doula to address this in practice?
- What feels uncertain or hard about this conversation for you?

An Orientation to Other Common Complications

Gestational Diabetes

Affects 6–9% of pregnancies. Managed with diet, exercise, sometimes insulin. Increases risk of macrosomia, induction, and cesarean.

Doula: Normalize the diagnosis. Support diet changes without judgment.

Preterm Labor

Contractions + cervical change before 37 weeks. Signs: regular contractions, pelvic pressure, low backache, watery discharge.

Doula: Encourage immediate evaluation. Do not wait and see.

Placenta Previa & Abruption

Previa: placenta covers cervix — vaginal birth not possible. Abruption: placenta separates — life-threatening emergency.

Doula: Painless vaginal bleeding in T3 = emergency evaluation NOW.

Hyperemesis Gravidarum

Severe, persistent nausea/vomiting requiring medical treatment. Not "just" morning sickness. May require IV fluids and hospitalization.

Doula: Validate the severity — it is often dismissed. Your presence matters.

What Would You Do?

The Scenario

Your client is 34 weeks pregnant. She texts you at 9:00 PM: "I have a terrible headache and my face looks puffy. It came on fast. Should I be worried?" Her sister had preeclampsia last year.

In your group, discuss:

- What do you text back — word for word?
- What would change if she said the headache started yesterday?
- She already called her provider and was told to "just rest and drink water" — what now?

WHAT IF she also mentions seeing spots? → 911 or immediate drive to L&D. Do not wait.

What to carry with you from tonight

01

A 40-week transformation. Understand it to support it.

02

Prenatal period is where trust is built.

03

Normalize discomforts. Evidence-based strategies are clinical skills.

04

Memorize warning signs. Preeclampsia can develop postpartum.

05

Maternal disparities are systemic. Your advocacy matters.

06

Your scope: observe, educate, advocate, support — never diagnose.